Multimodal Knee Osteoarthritis Detection Using Vibro-arthrographic Signals and X-ray Images

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CERTIFICATE

This is to certify that this project entitled **"MULTIMODAL KNEE OSTEOARTHRITIS DETECTION USING VIBRO-ARTHROGRAPHIC SIGNALS AND X-RAY IMAGES"** is the bonafied work carried out by **G. Shravani, N. Ramya Sree, G. Hari Chandhana, B. Soumya, V. Shashidhar,** as a Capstone Project for the partial fulfillment to award the degree **BACHELOR OF TECHNOLOGY** in **School of Computer Science and Artificial Intelligence** during the academic year 2025-2026 under our guidance and Supervision.

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Abstract

Knee osteoarthritis is a common degenerative joint disorder that affects mobility and quality of life, especially among elderly individuals and athletes. Early detection and accurate severity assessment are crucial for effective treatment and prevention of further damage. This project proposes a multimodal knee health assessment system that integrates vibroarthrographic (VAG) signal analysis and X-ray image-based classification to provide a comprehensive diagnosis.

The VAG signal module utilizes machine learning and deep learning models such as Support Vector Machine (SVM), Random Forest, and Convolutional Neural Networks (CNN) to classify knee conditions based on extracted signal features. The X-ray module employs a CNN-based architecture to classify osteoarthritis severity according to the Kellgren–Lawrence grading system (Grade 0–4). A decision-level fusion mechanism is implemented to combine the outputs of both modules, enabling improved diagnostic reliability.

The proposed system enhances early detection through functional analysis while providing structural validation through imaging. Experimental results demonstrate that the multimodal approach improves classification accuracy compared to single-modality systems. The system offers a non-invasive, cost-effective, and scalable solution for knee health monitoring, with potential applications in clinical diagnosis, telemedicine, and healthcare systems.

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1. Introduction

1.1 Background

Knee osteoarthritis (OA) is one of the most common degenerative joint diseases worldwide, particularly affecting elderly individuals and people involved in physically demanding activities. It leads to pain, stiffness, and reduced mobility, significantly impacting quality of life. Early diagnosis is essential to slow disease progression and improve treatment outcomes, but conventional diagnostic methods often fail to detect the condition in its initial stages.

Traditionally, knee osteoarthritis is diagnosed using radiographic imaging such as X-rays, which assess structural changes in the joint using the Kellgren–Lawrence grading system. While this method is widely accepted, it primarily identifies damage that has already occurred and may not reflect early functional abnormalities. As a result, there is a need for complementary techniques that can capture subtle changes before visible structural damage appears.

Vibroarthrography (VAG) is a non-invasive technique that records vibration signals produced by joint movement. These signals provide valuable information about the functional condition of the knee joint. Advances in signal processing and machine learning have enabled the use of VAG features to detect abnormalities and classify knee conditions. This approach offers a cost-effective and radiation-free alternative for early screening.

With the rapid development of artificial intelligence, integrating multiple data modalities has become a promising direction in healthcare diagnostics. Combining functional data from VAG signals with structural information from X-ray images can provide a more comprehensive understanding of knee health. This project leverages this multimodal approach to improve diagnostic accuracy and support early detection and severity assessment of knee osteoarthritis.

1.2 Objectives

1.2.1 Core Objectives

The primary objective of this research is to design and develop a multimodal AI-based knee health assessment system capable of accurately detecting knee conditions and assessing osteoarthritis severity. The system focuses on leveraging machine learning, deep learning, and multimodal data fusion techniques to improve diagnostic accuracy and enable early detection. The core objectives include:

• **VAG Signal Analysis:** To analyse vibro-arthrographic (VAG) signal features such as RMS amplitude, spectral entropy, and frequency components to classify knee conditions (normal, osteoarthritis, and ligament injury) using machine learning and deep learning models.

• **X-ray Based Severity Classification:** To develop a convolutional neural network (CNN) model for classifying knee osteoarthritis severity based on X-ray images using the Kellgren–Lawrence grading system (Grade 0–4).

• **Multimodal Fusion:** To integrate functional information from VAG signals and structural information from X-ray images using decision-level fusion techniques for improved diagnostic reliability.

• **Model Evaluation and Optimization:** To evaluate the performance of different machine learning and deep learning models using metrics such as accuracy, precision, recall, and F1-score, and optimize them for better results.

• **User Interface Development:** To design a user-friendly interface that allows easy input of VAG data and X-ray images and displays clear diagnostic outputs including condition, severity, and recommendations.

1.2.2 Advanced Objectives

The advanced objective of this research is to enhance the proposed multimodal knee health assessment system by incorporating intelligent data fusion, improved model performance, and real-world applicability. The system aims to extend beyond basic classification by providing a more robust, scalable, and clinically relevant solution. The advanced objectives include:

• **Multimodal Data Fusion Optimization:** To implement and improve decision-level fusion techniques that effectively combine VAG signal predictions and X-ray severity outputs for more accurate and reliable diagnosis.

• **Comparative Model Analysis:** To perform a detailed comparison between classical machine learning models and deep learning approaches to identify the most efficient and accurate model for biomedical signal classification.

• **System Scalability and Deployment:** To design the system in a scalable manner that supports cloud-based deployment and integration with IoT wearable devices for real-time knee health monitoring.

• **Real-world Application Readiness:** To extend the system for practical healthcare usage by enabling continuous monitoring, remote diagnosis, and integration with telemedicine platforms.

1.3 Scope of the Project

1.3.1 In-Scope Features

The scope of this project includes the design and development of a multimodal knee health assessment system that integrates vibroarthrographic (VAG) signal analysis and X-ray image classification. The system focuses on providing early detection and severity assessment of knee osteoarthritis using artificial intelligence techniques. The in-scope features of the project are as follows:

- **VAG Signal-Based Classification:** The system processes pre-extracted VAG signal features to classify knee conditions such as normal, osteoarthritis, and ligament injury using machine learning and deep learning models.
- **X-ray Image-Based Severity Analysis:** The system utilizes convolutional neural networks (CNN) to classify knee osteoarthritis severity based on the Kellgren–Lawrence grading system (Grade 0–4).
- **Data Preprocessing and Feature Engineering:** Includes normalization, scaling, encoding of VAG features, and image preprocessing such as resizing and augmentation for X-ray data.
- **Multimodal Decision-Level Fusion:** Integration of outputs from both VAG and X-ray modules to generate a comprehensive diagnosis.
- **Model Evaluation and Performance Analysis:** Implementation of evaluation metrics such as accuracy, precision, recall, F1-score, and confusion matrix for performance comparison.
- **User Interface Prototype:** A basic interface for uploading VAG data and X-ray images and displaying prediction results in a structured format.
- **Result Visualization:** Display of final diagnosis including knee condition, severity level, and suggested interpretation.

- **Local Authentication System:** Implementation of a basic login and registration module, with future extensibility to connect to MySQL or other database systems.

1.3.2 Data and Machine Learning Scope

The data and machine learning scope of this project focuses on utilizing structured VAG signal features and unstructured X-ray image data to develop accurate and efficient predictive models for knee health assessment. The system operates on publicly available datasets, ensuring accessibility and reproducibility of results.

The VAG dataset consists of pre-extracted numerical features such as RMS amplitude, spectral entropy, peak frequency, zero-crossing rate, and mean frequency. These features are used to train machine learning models including Support Vector Machine (SVM), Random Forest, and deep learning models for classification of knee conditions.

The X-ray dataset consists of labeled medical images categorized according to the Kellgren–Lawrence grading system. These images are processed using convolutional neural networks (CNN) with image preprocessing techniques such as resizing, normalization, and augmentation to improve model performance.

The machine learning scope includes data preprocessing, feature scaling, model training, validation, and evaluation using standard performance metrics such as accuracy, precision, recall, F1-score, and confusion matrix. Comparative analysis between different models is performed to select the most suitable approach.

Additionally, the system incorporates a decision-level fusion mechanism to combine outputs from both modalities, enhancing prediction accuracy and reliability. The scope is limited to supervised learning approaches and does not include real-time streaming data or unsupervised learning techniques.

1.3.3 Out-of-Scope Features

The project focuses on developing a multimodal AI-based knee health assessment system; however, certain functionalities are beyond the scope of this implementation.

- **Real-time Data Acquisition:** The system does not include live collection of VAG signals using hardware sensors or real-time streaming from wearable devices. It operates only on pre-collected datasets.

- **Clinical Validation:** The system is not validated with real patient clinical trials or hospital-grade datasets and is intended for research and academic purposes only.
- **Automated Treatment Recommendation System:** The project does not provide personalized treatment plans, medication suggestions, or medical prescriptions.
- **Integration with Hospital Systems:** The system is not integrated with electronic health records (EHR), hospital databases, or healthcare management systems.
- **Advanced Explainable AI Techniques:** While basic prediction outputs are provided, advanced explainability methods such as Grad-CAM or SHAP visualizations are not implemented.
- **Mobile or Web Deployment at Production Scale:** The system includes only a prototype interface and is not deployed as a fully functional mobile or web application.
- **Other Joint or Disease Analysis:** The scope is limited to knee-related conditions and does not extend to other joints or orthopedic diseases.

1.4 Significance of the Project

The proposed Multimodal Knee Health Assessment System holds substantial significance in the field of healthcare by addressing critical limitations associated with traditional diagnostic approaches for knee osteoarthritis. Knee osteoarthritis is a widespread degenerative condition that affects millions of individuals globally, leading to pain, reduced mobility, and diminished quality of life. Early detection of knee disorders is essential for initiating timely treatment, preventing further joint deterioration, and improving long-term patient outcomes. However, conventional diagnostic techniques primarily rely on radiographic imaging such as X-rays, which are effective only in identifying structural damage at later stages. This creates a gap in early-stage diagnosis, where functional abnormalities remain undetected.

This project introduces an innovative and integrated solution by combining vibro-arthrographic (VAG) signal analysis with X-ray image classification. The use of VAG signals enables the detection of subtle joint vibrations and irregular movement patterns, which can indicate early functional abnormalities before structural damage becomes visible. At the same time, X-ray imaging provides a clear understanding of the physical condition and severity of joint degradation. By integrating these two complementary modalities, the system achieves a more comprehensive

and accurate assessment of knee health, addressing both functional and structural aspects simultaneously.

The incorporation of machine learning and deep learning techniques further enhances the diagnostic capability of the system. Advanced algorithms are capable of identifying complex patterns within both signal and image data, leading to improved accuracy, consistency, and reliability compared to manual analysis. This reduces the dependency on subjective interpretation by medical professionals and minimizes the chances of diagnostic variability. Additionally, the system supports early-stage detection, which plays a crucial role in preventive healthcare by enabling timely intervention and reducing the need for invasive treatments or surgeries.

Another important aspect of the project is its focus on non-invasive and cost-effective diagnostic methods. Unlike traditional approaches that may require expensive imaging equipment or clinical procedures, VAG signal analysis offers a low-cost alternative for preliminary screening. This makes the system particularly beneficial for deployment in resource-limited settings such as rural and remote areas, where access to advanced medical facilities is limited. The ability to provide accurate and accessible diagnostic support can significantly improve healthcare delivery in underserved regions.

Furthermore, the modular and scalable design of the system ensures flexibility for future enhancements. The system can be extended to integrate real-time data acquisition through wearable IoT devices, enabling continuous monitoring of knee health. It also has the potential to be incorporated into telemedicine platforms, allowing remote diagnosis and consultation. These capabilities align with the growing trend of digital healthcare and smart medical systems.
In conclusion, the proposed system represents a significant advancement in AI-driven healthcare solutions by providing a reliable, efficient, and accessible tool for knee health assessment. Its ability to combine multiple data sources, enhance diagnostic accuracy, and support early intervention makes it highly valuable for both clinical applications and future research in intelligent medical systems.

Industry Significance

The proposed Multimodal Knee Health Assessment System has strong relevance in the healthcare and medical technology industry. With the increasing prevalence of knee osteoarthritis due to aging populations, sedentary lifestyles, and sports injuries, there is a growing demand for efficient, early-stage diagnostic solutions. This project addresses this need by combining AI-driven analysis with non-invasive techniques, aligning with current industry trends in digital health and precision medicine.

The system supports the advancement of **AI in healthcare diagnostics**, enabling faster and more accurate assessment compared to traditional manual methods. It can be adopted by hospitals, diagnostic centres, physiotherapy clinics, and telemedicine platforms to improve patient screening and reduce diagnostic workload. The integration of VAG signals introduces a cost-effective alternative for early detection, reducing dependency on expensive imaging systems.

From an industry perspective, the solution has potential for commercialization through SaaS platforms, integration with wearable IoT devices, and deployment in remote healthcare environments. It also aligns with the global shift towards preventive healthcare and remote monitoring systems. Overall, the project contributes to the development of intelligent, scalable, and accessible healthcare technologies, making it highly relevant for the medical AI and health-tech industry.

1.5 Challenges and Limitations

Challenges

The development of this system involves several technical and functional challenges that impact the accuracy and efficiency of its operations:

- **Multimodal Data Integration:** Combining VAG signal data and X-ray images is challenging due to differences in data formats, requiring careful preprocessing and fusion techniques to ensure consistent and accurate predictions.
- **Imperfect Fuzzy Matching:** **Feature Extraction Complexity:** Extracting meaningful features from VAG signals and identifying relevant patterns in X-ray images requires proper preprocessing, normalization, and model tuning to avoid noise and inaccuracies.

- **Model Selection and Optimization:** Choosing the most suitable machine learning and deep learning models (SVM, Random Forest, CNN) and tuning hyperparameters for optimal performance is a complex and time-consuming task.
- **Class Imbalance in Dataset:** Uneven distribution of classes (normal vs diseased cases) can affect model performance and lead to biased predictions if not handled properly.
- **Overfitting in Deep Learning Models:** Training CNN models on limited datasets may result in overfitting, reducing the model's ability to generalize to new data.
- **Fusion Strategy Design:** Developing an effective decision-level fusion mechanism to combine outputs from both modules without introducing errors is challenging.
- **Computational Constraints:** Training deep learning models requires sufficient computational resources, and limited hardware availability may affect performance and training time.

Limitations

Despite the system's utility and features, several limitations restrict its scope and performance:

- **Dataset Dependency: Use of Synthetic/Public Data:** The system is trained on publicly available datasets, which may not fully represent real-world clinical conditions.
- **Lack of Clinical Validation:** The model has not been validated with real patient data or tested in hospital environments.
- **No Real-Time Data Acquisition:** The system does not include live VAG signal collection using sensors and relies only on pre-existing data.
- **Limited Explainability:** Advanced explainable AI techniques (e.g., heatmaps, SHAP) are not fully implemented, limiting interpretability for medical professionals.
- **Prototype-Level Implementation:** The system is developed as a prototype and is not deployed as a full-scale application.
- **No Treatment Recommendation System:** The system provides diagnosis only and does not generate personalized treatment plans or medical advice.
- **Limited Scope:** The project focuses only on knee-related conditions and does not extend to other joints or diseases.

2. Related Work

The paper by Piccolo et al [1]. provides a comprehensive review of imaging modalities used in the diagnosis and evaluation of knee osteoarthritis (KOA), highlighting the strengths and limitations of each technique within a multimodal framework. The authors emphasize that conventional radiography remains the first-line and most widely used method due to its cost-effectiveness and accessibility, although it is limited in detecting early-stage and soft tissue changes. Advanced imaging techniques such as MRI are identified as the most sensitive tools for assessing cartilage, synovium, and other joint structures, enabling early detection of biochemical and morphological alterations. disease grading, and treatment planning. Overall, this review underscores the importance of integrating multiple imaging approaches along with advanced technologies to achieve more accurate diagnosis and better management of KOA.

The The paper by Wang et al [2]. (2025) presents a novel, non-invasive approach for the detection of knee osteoarthritis (OA) using vibroarthrographic (VAG) signals combined with deep learning techniques. The authors propose a dual-model framework employing both 1D and 2D convolutional neural networks (CNNs) to analyze time-domain and frequency-domain representations of knee joint vibrations, respectively. Using data collected from 128 subjects through accelerometer-based sensors placed at multiple knee joint locations, the study demonstrates that the 2D CNN (MobileNet-based) achieves higher classification accuracy (0.764 ± 0.010), while the lightweight 1D CNN offers advantages for real-time edge computing due to its low memory footprint (234 KB). Despite promising results, the study also notes challenges related to model generalization, class imbalance, and variability across joint positions. Overall, this work contributes to the growing body of research supporting AI-driven, wearable, and non-invasive diagnostic systems for improved clinical assessment of knee osteoarthritis.

The The study by Leuthard et al [3]. proposes an innovative approach for the early detection and monitoring of knee osteoarthritis using acoustic emission (AE) biomarkers. Recognizing the limitations of conventional diagnostic methods such as radiography, which often fail to detect early-stage osteoarthritis, the authors emphasize the need for noninvasive, cost-effective, and sensitive diagnostic tools. The paper outlines a prospective, single-center exploratory study that

integrates acoustic, kinematic, and thermal data collected a wearable knee brace system, alongside gold-standard measures like MRI and Kellgren-Lawrence grading for validation. By employing advanced signal processing and machine learning techniques. Although the study acknowledges limitations such as sample size and population specificity, it significantly contributes to the growing body of research supporting acoustic analysis as a promising, radiation-free alternative for early osteoarthritis diagnosis and improved clinical decision-making.

The study by Vatolik et al [4]. presents an innovative approach for monitoring knee joint conditions through the development of a multi-modal sensor network that integrates acoustic emission (AE), kinematic, and dynamic data. The authors address the limitations of traditional imaging techniques, such as MRI and X-rays, which, despite their accuracy, are often costly and impractical for continuous or home-based monitoring. By utilizing a sensitive microphone-based system to capture acoustic signals generated during joint movement, the study demonstrates that distinct acoustic patterns can differentiate between healthy and osteoarthritic knees. Although the study is limited by a small sample size, it highlights the feasibility of developing portable, cost-effective diagnostic tools and emphasizes the need for further research to validate and standardize this approach for clinical applications.

A comprehensive review [5] of existing and emerging diagnostic approaches for knee osteoarthritis (KOA), highlighting the limitations of conventional imaging techniques such as MRI, radiography, and arthroscopy. Although these methods allow direct visualization of joint structures, the study emphasizes their poor correlation with clinical symptoms, which restricts their effectiveness in early diagnosis and disease monitoring the authors discuss the growing interest in non- or minimally invasive diagnostic alternatives, including biochemical markers and acoustic emission (AE) analysis, which aim to capture functional and physiological changes in joint tissues. While biochemical markers offer insight into cartilage degradation and inflammation, their clinical Overall, this review underscores the need for reliable, non-invasive, and data-driven diagnostic tools to advance personalized medicine in KOA management.

The The paper by Al-Wasti (2025) [6] provides a comprehensive review of non-invasive vibration-based techniques for assessing tibial mechanical properties, highlighting their growing importance

in orthopaedic diagnostics and rehabilitation. The study emphasizes that conventional methods such as DXA and imaging techniques, although useful, are limited by factors like radiation exposure and inability to detect early structural changes. Overall, the study concludes that vibration-based assessment holds strong potential for personalized and continuous monitoring of bone health, while recommending further research focused on standardization, validation, and integration with other diagnostic modalities to enhance its effectiveness in clinical practice.

The study presents a novel [7] and effective deep learning–based approach for automated detection and classification of knee osteoarthritis (OA) using X-ray images. It highlights the limitations of traditional diagnostic methods, such as manual interpretation using the Kellgren–Lawrence (KL) grading system, which can be subjective and time-consuming. To address these challenges, the authors propose a hybrid model that integrates VGG16 and Convolutional Neural Network (CNN) architectures, leveraging their complementary strengths in feature extraction and spatial analysis. The study conducts a comparative analysis with other state-of-the-art models, including VGG19 and ResNet50, and demonstrates that the proposed model achieves superior performance, with accuracy exceeding 93% across all OA stages while also identifying scope for future improvements such as multi-modal integration and enhanced clinical validation.

The reviewed study presents [8] a significant advancement in the automated diagnosis of knee osteoarthritis (OA) through the application of deep learning techniques. Prior research in this domain has explored various machine learning and deep learning approaches, including CNNs, FCNs, YOLO-based models, The study also incorporates data augmentation techniques to address class imbalance, thereby improving model generalization and reliability. Through comprehensive comparative analysis with models such as VGG19 and ResNet50, the proposed approach demonstrates superior performance, achieving an accuracy exceeding 93% across all OA stages. Overall, the paper contributes a robust and efficient computer-aided diagnostic framework, emphasizing the importance of hybrid architectures and balanced datasets while suggesting future scope for multi-modal integration and clinical validation.

The study proposes a novel [9] piezoresistive-based gait monitoring technique for the recognition of knee osteoarthritis (OA) patients, addressing the limitations of traditional imaging methods that

fail to capture dynamic joint function and are unsuitable for long-term monitoring. The authors develop a wearable insole system embedded with a 48-channel piezoresistive sensor array to capture plantar pressure signals during walking, particularly Support Vector Machine (SVM), for classification. The proposed model achieves high performance, with an accuracy of 93.15%, demonstrating its effectiveness in distinguishing OA patients from healthy individuals. Furthermore, the study highlights the advantages of the system, including portability, low cost, and suitability for remote health monitoring. Overall, the paper contributes significantly to the field of intelligent healthcare by introducing a practical and efficient wearable solution for early detection and monitoring of knee osteoarthritis.

The study [10] presents an innovative wearable-based approach for the recognition of knee osteoarthritis (OA) using a piezoresistive gait monitoring system. Unlike conventional diagnostic techniques that rely on static imaging and fail to capture dynamic joint behaviour, the proposed method utilizes plantar pressure signals obtained from a 48-channel sensor-embedded insole to analyse gait patterns during walking. By extracting biomechanical features such as contact time and maximum force across different foot regions, the study employs machine learning algorithms, particularly Support Vector Machine (SVM), achieving a high classification accuracy of 93.15%, along with strong precision and recall values. which may affect generalizability. Overall, the paper contributes significantly to intelligent healthcare by introducing a practical, accurate, and non-invasive solution for OA detection, while emphasizing the need for further validation and enhancement for real-world clinical applications.

3. Problem Statement

In Knee osteoarthritis Knee osteoarthritis (OA) is a progressive degenerative joint disorder that significantly affects mobility, flexibility, and overall quality of life, particularly among elderly individuals, athletes, and people with physically demanding lifestyles. The condition leads to cartilage degradation, joint pain, stiffness, and reduced movement, making daily activities increasingly difficult. As the disease progresses, it can result in severe disability if not diagnosed and managed at an early stage. Therefore, early detection and accurate assessment of disease severity are crucial for effective treatment planning, rehabilitation, and prevention of further joint deterioration.

Traditional diagnostic methods for knee osteoarthritis primarily rely on radiographic imaging techniques such as X-rays, which evaluate structural changes in the joint using the Kellgren–Lawrence grading system. While X-ray imaging is widely accepted in clinical practice, it mainly detects visible structural damage that occurs in later stages of the disease. As a result, early functional abnormalities, such as subtle joint vibrations and irregular movement patterns, often go unnoticed. This limitation reduces the effectiveness of early intervention strategies and delays appropriate medical response.

Furthermore, conventional diagnostic approaches depend heavily on manual interpretation by medical professionals, which can be time-consuming and subject to inter-observer variability. Differences in expertise and experience among clinicians may lead to inconsistent assessments. In addition, existing systems often rely on a single type of data (either imaging or clinical observation), lacking the ability to integrate multiple data sources for a more comprehensive evaluation of knee health.

To overcome these challenges, there is a growing need for intelligent and automated diagnostic systems that can analyse both functional and structural aspects of the knee joint. This project proposes a multimodal AI-based solution that combines vibro-arthrographic (VAG) signal analysis with X-ray image classification. VAG signals capture joint vibrations during movement, enabling early detection of abnormalities, while X-ray images provide structural insights into disease severity.

By integrating machine learning techniques for signal analysis and deep learning models for image classification, the proposed system enhances diagnostic accuracy and reliability. The inclusion of a decision-level fusion mechanism further improves performance by combining insights from both

modalities. This approach not only supports early detection but also enables more precise severity assessment. Ultimately, the system aims to serve as a cost-effective, non-invasive, and scalable decision-support tool for healthcare professionals, contributing to improved patient outcomes and advancing AI-driven medical diagnostics.

4. Requirement Analysis

4.1 Functional Requirements

The functional requirements define the core operations and features that the proposed system must perform to achieve its objectives. The following functionalities are included in the system:

• **User Input Handling:** The system should allow users to upload VAG signal data (CSV format) and knee X-ray images through the interface.

• **Data Preprocessing:** The system must preprocess input data, including feature scaling and encoding for VAG data, and resizing, normalization, and augmentation for X-ray images.

• **VAG-Based Classification:** The system should analyze VAG signal features and classify knee conditions such as normal, osteoarthritis, and ligament injury using machine learning/deep learning models.

• **X-ray-Based Severity Classification:** The system must classify osteoarthritis severity (Grade 0–4) from X-ray images using a CNN model.

• **Multimodal Fusion:** The system should integrate outputs from both VAG and X-ray modules using decision-level fusion to generate a final diagnosis.

• **Result Generation:** The system must display the final output including knee condition, severity level, and interpretation in a clear and structured format.

• **Performance Evaluation:** The system should evaluate model performance using metrics such as accuracy, precision, recall, F1-score, and confusion matrix.

• **User Interface Interaction:** The system should provide a simple and user-friendly interface for data input, processing, and result visualization.

4.2 Non-Functional Requirements

Non-functional requirements define the quality attributes and constraints of the system to ensure efficient, reliable, and user-friendly operation.

• **Performance:** The system should provide fast and efficient processing of VAG data and X-ray images with minimal response time.

• **Usability:** The interface should be simple, intuitive, and easy to use for both technical and non-technical users.

• **Scalability:** The system should be designed to handle increasing data volume and support future enhancements such as real-time data and additional modules.

• **Reliability:** The system should produce consistent and accurate results with minimal errors during operation.

• **Maintainability:** The system should be modular and easy to update, debug, and improve without affecting overall functionality.

• **Security & Privacy:** The system should ensure that user data is handled securely and does not store sensitive information unnecessarily.

• **Portability:** The system should be able to run on different platforms such as local machines or cloud environments.

• **Availability:** The system should be accessible whenever required, especially when deployed on cloud platforms.

5. Risk Analysis

The development and deployment of the this system involve several potential risks that must be identified and managed to ensure reliable performance, user satisfaction, and long-term sustainability. The key risks include:

Risk Category	Description	Impact	Likelihood	Mitigation Strategy
Data Quality Risk	Public datasets may not represent real clinical conditions	High	Medium	Use diverse datasets and apply proper preprocessing & validation
Model Performance Risk	Models may overfit, underfit, or misclassify data	High	Medium	Perform hyperparameter tuning and cross-validation
Integration Risk	Difficulty in combining VAG and X-ray outputs	Medium	Medium	Thorough testing of fusion logic and validation
Computational Risk	Limited hardware resources for training deep models	Medium	Medium	Use cloud platforms like Google Colab

Risk Category	Description	Impact	Likelihood	Mitigation Strategy
Input Data Risk	Poor quality input (e.g., unclear X-rays) affects results	High	High	Implement input validation and user guidance
Deployment Risk	Challenges in scaling or deploying system in real-world	Medium	Low	Use modular and scalable system design
Ethical/Legal Risk	Misuse or misinterpretation of results	High	Medium	Clearly state system limitations and usage guidelines

6. Feasibility Analysis

6.1 Technical Feasibility

The proposed Multimodal Knee Health Assessment System is technically feasible due to the availability of established machine learning and deep learning frameworks, publicly accessible datasets, and scalable computing platforms. The system utilizes widely used technologies such as Python, TensorFlow, Scikit-learn, and OpenCV, which are well-supported and suitable for implementing both VAG signal analysis and X-ray image classification.

The VAG dataset consists of structured numerical features, which can be efficiently processed using classical machine learning models such as Support Vector Machine (SVM) and Random Forest, as well as deep learning models. The X-ray dataset is suitable for convolutional neural network (CNN) based image classification, supported by transfer learning techniques for improved performance.

Additionally, cloud-based platforms like Google Colab provide sufficient computational resources, eliminating the need for high-cost hardware. The modular architecture of the system ensures easy integration of different components and supports scalability for future enhancements. Overall, the required tools, technologies, and data resources are readily available, making the development and implementation of the system technically achievable within the given project scope.

6.2 Operational Feasibility

The system is operationally feasible as it is designed with a simple and user-friendly interface, allowing users to easily upload VAG data and X-ray images and obtain results without requiring advanced technical knowledge. The workflow is straightforward, involving data input, processing, and result generation.

The system can be used by healthcare professionals, researchers, and students for analysis and learning purposes. It requires minimal training to operate and does not depend on complex infrastructure. Additionally, the modular design ensures easy maintenance and future enhancements.

Overall, the system is practical, easy to use, and can be effectively implemented in real-world scenarios such as research labs, clinics, and educational institutions

6.3 Economic Feasibility

The proposed system is economically feasible as it primarily relies on free and open-source tools such as Python, TensorFlow, Scikit-learn, and OpenCV, eliminating software licensing costs. The datasets used are publicly available and free for research purposes. Cloud platforms like Google Colab can be utilized for model training without requiring expensive hardware infrastructure.

The only minimal expenses involved include internet usage, basic system maintenance, and documentation costs, which are negligible in an academic setup. Since no specialized equipment or paid services are required, the overall development cost remains low. This makes the project cost-effective and suitable for implementation in educational environments as well as potential scaling in real-world applications.

7. Proposed Solution

The The proposed system presents a comprehensive multimodal AI-based knee health assessment framework designed to improve the early detection and severity evaluation of knee osteoarthritis. The system integrates vibroarthrographic (VAG) signal analysis with X-ray image classification, thereby combining both functional and structural aspects of knee joint assessment. This multimodal approach addresses the limitations of traditional single-modality diagnostic systems by providing a more holistic and reliable evaluation of knee health.

The system architecture is organized into three primary modules, each responsible for a specific stage of data processing and analysis. **Module 1 focuses on VAG signal analysis**, where pre-extracted features such as Root Mean Square (RMS) amplitude, spectral entropy, peak frequency, zero-crossing rate, and mean frequency are utilized. These features represent the vibrational characteristics of the knee joint during movement and provide valuable insights into functional abnormalities. The processed data is fed into various machine learning and deep learning models, including Support Vector Machine (SVM), Random Forest, and Convolutional Neural Networks (CNN). These models are trained to classify knee conditions into categories such as normal, osteoarthritis, and ligament injury. This module plays a crucial role in identifying early-stage abnormalities that may not be visible through imaging techniques.

Module 2 is dedicated to X-ray image-based severity classification, where advanced deep learning architectures such as ResNet50, DenseNet121, EfficientNet-B0, and a custom CNN model are employed. These models utilize transfer learning techniques, allowing them to leverage pre-trained weights from large-scale image datasets and adapt them to the specific task of medical image classification. The X-ray images are preprocessed through resizing, normalization, and augmentation to enhance model performance and generalization. The models are trained to classify osteoarthritis severity based on the Kellgren–Lawrence grading system, which ranges from Grade 0 (normal) to Grade 4 (severe). Among these models, ResNet50 demonstrated superior performance, achieving the highest accuracy, thereby validating the effectiveness of deep feature extraction in medical imaging.

Module 3 implements a decision-level fusion mechanism, which serves as the integration layer of the system. In this module, the outputs from both VAG signal analysis and X-ray classification are combined to generate a final diagnosis. The fusion process ensures that both functional insights (from VAG signals) and structural information (from X-rays) are considered simultaneously, resulting in improved diagnostic accuracy and reliability. This approach reduces the risk of misclassification and enhances confidence in the system's predictions.

The overall workflow of the system includes multiple stages, starting from user input, where VAG data and/or X-ray images are uploaded. This is followed by data preprocessing, feature extraction, model prediction, and multimodal fusion. The final output is presented through a user-friendly interface, displaying the predicted knee condition, severity level, and a basic interpretation of the results.

The proposed system offers a cost-effective, non-invasive, and scalable solution for knee health assessment. It has the potential to assist healthcare professionals in early diagnosis, support clinical decision-making, and enable preventive healthcare strategies. Furthermore, the modular design of the system allows for future enhancements such as real-time monitoring, integration with wearable devices, and deployment in telemedicine platforms, making it a valuable contribution to AI-driven healthcare solutions.

8. Architecture Diagram & Flow Chart

Architecture Diagram

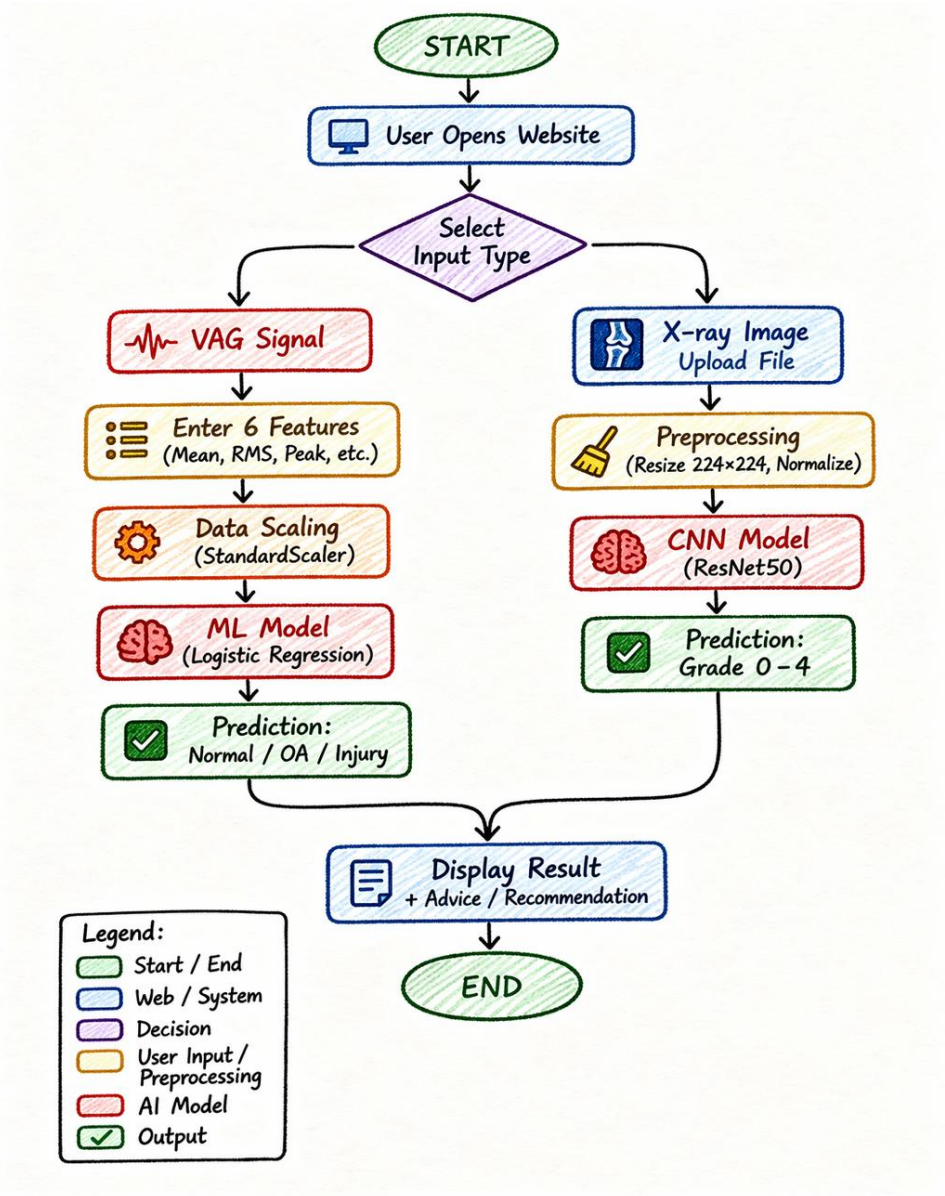

Fig. 1

Flowchart

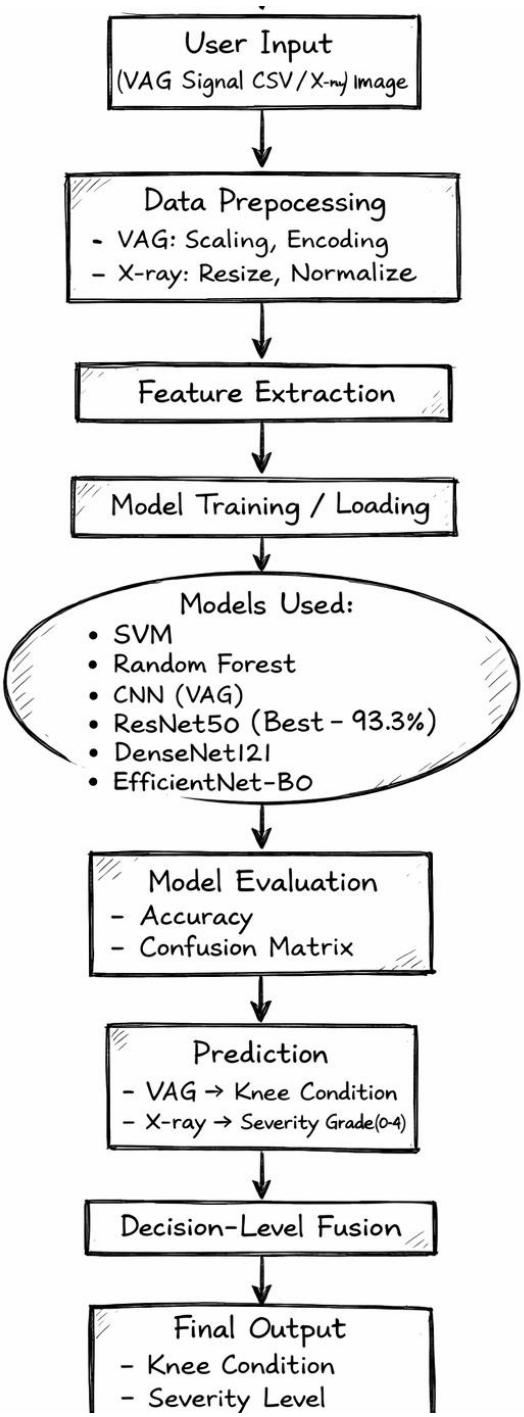

Fig.2

9. Simulation set up and Implementation

The simulation and implementation of the Multimodal Knee Health Assessment System were designed to evaluate the performance, accuracy, and real-world applicability of an AI-driven diagnostic pipeline. The system integrates vibroarthrographic (VAG) signal analysis and X-ray image classification using machine learning and deep learning techniques. The implementation follows a modular architecture consisting of data preprocessing, model training, prediction, and multimodal fusion, ensuring scalability and efficiency.

9.1 Simulation Environment Setup

a) Hardware Configuration

The system was developed and tested using standard computing resources to ensure accessibility and feasibility in academic and low-resource environments. The hardware configuration includes:

• Processor: Intel Core i5/i7 or AMD equivalent for efficient computation

• RAM: Minimum 8 GB (16 GB recommended for smoother deep learning operations)

• Storage: Approximately 2–5 GB for datasets, trained models, and outputs

• GPU: Optional; GPU acceleration (via Google Colab) was used for faster training of deep learning models such as ResNet50 and EfficientNetB0

The system is optimized to run efficiently even on CPU-based systems, ensuring wider usability.

b) Software Requirements

Software Requirements	Details
Programming Language	Python 3.8+
Development Platform	Google Colab / Jupyter Notebook
ML Libraries	Scikit-learn
DL Libraries	TensorFlow, Keras
Data Processing	NumPy, Pandas
Visualization	Matplotlib, Seaborn

Software Requirements	Details
Image Processing	OpenCV
Model Saving	pickle, Keras (.keras format)
UI Development	Streamlit / Basic Python Interface

The implementation uses open-source tools, making the system cost-effective and easy to reproduce.

The system follows a **multi-stage processing pipeline**, where raw input data is transformed into meaningful predictions through structured processing steps. It supports both structured (VAG features) and unstructured (X-ray images) data.

9.2 Data Acquisition and Preprocessing

a) VAG Dataset Preparation

The VAG dataset contains numerical features representing knee joint vibrations. Preprocessing steps include:

• Removal of missing or inconsistent values

• Encoding categorical attributes such as knee_condition and severity_level

• Feature scaling using StandardScaler to normalize data distribution

• Selection of key features such as RMS amplitude, spectral entropy, peak frequency, zero-crossing rate, and mean frequency

These steps ensure that the dataset is clean, normalized, and suitable for machine learning models.

b) X-ray Dataset Preparation

The X-ray dataset consists of labeled medical images categorized according to osteoarthritis severity (KL grades 0–4). Preprocessing includes:

• Resizing all images to a uniform size (224×224 pixels)

• Normalizing pixel values to improve model convergence

• Applying data augmentation techniques such as rotation, flipping, and zooming to increase

dataset diversity

• Splitting the dataset into training, validation, and testing sets

This preprocessing improves generalization and reduces overfitting in deep learning models.

9.3 VAG Signal Classification (Module 1)

a) Machine Learning Models

The system implements classical machine learning models:

• Support Vector Machine (SVM) for high-dimensional feature classification

• Random Forest for robust ensemble learning

b) Deep Learning Model

• A CNN-based model is also implemented to capture non-linear relationships in VAG features

c) Implementation Workflow

• Data splitting into training and testing sets

• Model training using labeled data

• Performance evaluation using accuracy, precision, recall, and confusion matrix

This module focuses on identifying the functional condition of the knee joint.

9.4 X-ray Image Classification (Module 2)

a) Deep Learning Models Used

The following architectures were implemented using transfer learning:

• ResNet50 (Best performing model – 93.3% accuracy)

• DenseNet121

• EfficientNet-B0

• Custom CNN

b) Training Process

• Pre-trained models were fine-tuned on the dataset

• Final layers were modified for classification into 5 classes (Grade 0–4)

• Training was performed for multiple epochs (e.g., 50 epochs)

• Loss and accuracy curves were analyzed to monitor performance

c) Output

• Classification of osteoarthritis severity based on X-ray images

This module provides structural assessment of knee damage.

9.5 Multimodal Fusion (Module 3)

The system integrates outputs from both modules using **decision-level fusion**:

• VAG module predicts knee condition

• X-ray module predicts severity grade

• Fusion logic combines both outputs to produce a comprehensive diagnosis

This approach improves reliability by considering both functional and structural aspects.

9.6 User Interface Implementation

The system includes a simple and user-friendly interface:

User Functionalities

• Upload VAG CSV file

• Upload X-ray image

• View prediction results

Output Display

• Knee condition (Normal / Osteoarthritis / Ligament Injury)

• Severity level (Grade 0–4)

• Basic recommendation or interpretation

The UI ensures ease of use for both technical and non-technical users.

9.7 End-to-End System Simulation

Step 1: User Interaction

User uploads VAG data or X-ray image

Step 2: Backend Processing

• Data preprocessing

• Feature extraction

Step 3: Model Execution

• VAG classification

• X-ray classification

• Decision-level fusion

Step 4: Output Generation

• Final diagnosis displayed to the user

9.8 Validation and Testing

Functional Testing

• Verification of correct model predictions

• Validation of classification outputs

Performance Testing

• VAG model prediction time: < 1 second

• X-ray model prediction time: 1–3 seconds

Usability Testing

• Interface responsiveness

• Ease of navigation

• Clarity of output

9.9 Summary

The simulation demonstrates that the system effectively integrates machine learning and deep learning techniques to provide accurate knee health assessment. The multimodal approach enhances diagnostic reliability by combining functional and structural analysis. Among all models, **ResNet50 achieved the highest accuracy of 93.3%**, proving its effectiveness for medical image classification.

The system is efficient, scalable, and suitable for future real-world deployment in healthcare applications.

10. Results

RESNET50

Classification Report:

	precision	recall	f1-score	support
0Normal	0.93	0.94	0.94	268
1Doubtful	0.86	0.89	0.88	214
2Mild	0.86	0.80	0.83	119
3Moderate	0.96	0.92	0.94	113
4Severe	0.91	0.93	0.92	111
accuracy			0.90	825
macro avg	0.90	0.90	0.90	825
weighted avg	0.90	0.90	0.90	825

Fig.3

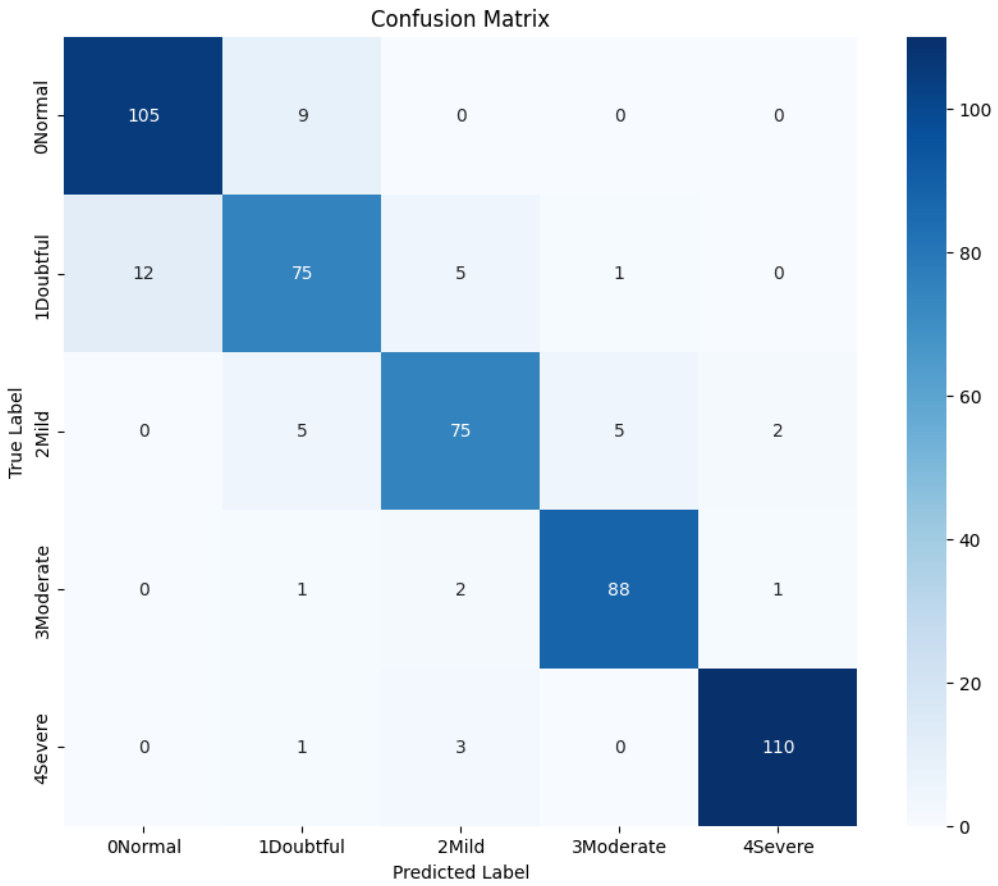

Fig.4

CNN

	precision	recall	f1-score	support
0	0.8421	0.8631	0.8525	241
1	0.8110	0.8766	0.8425	235
2	0.9082	0.7236	0.8054	123
3	0.8852	0.8504	0.8675	127
4	0.8269	0.8687	0.8473	99
accuracy			0.8448	825
macro avg	0.8547	0.8365	0.8430	825
weighted avg	0.8479	0.8448	0.8443	825

Fig.5

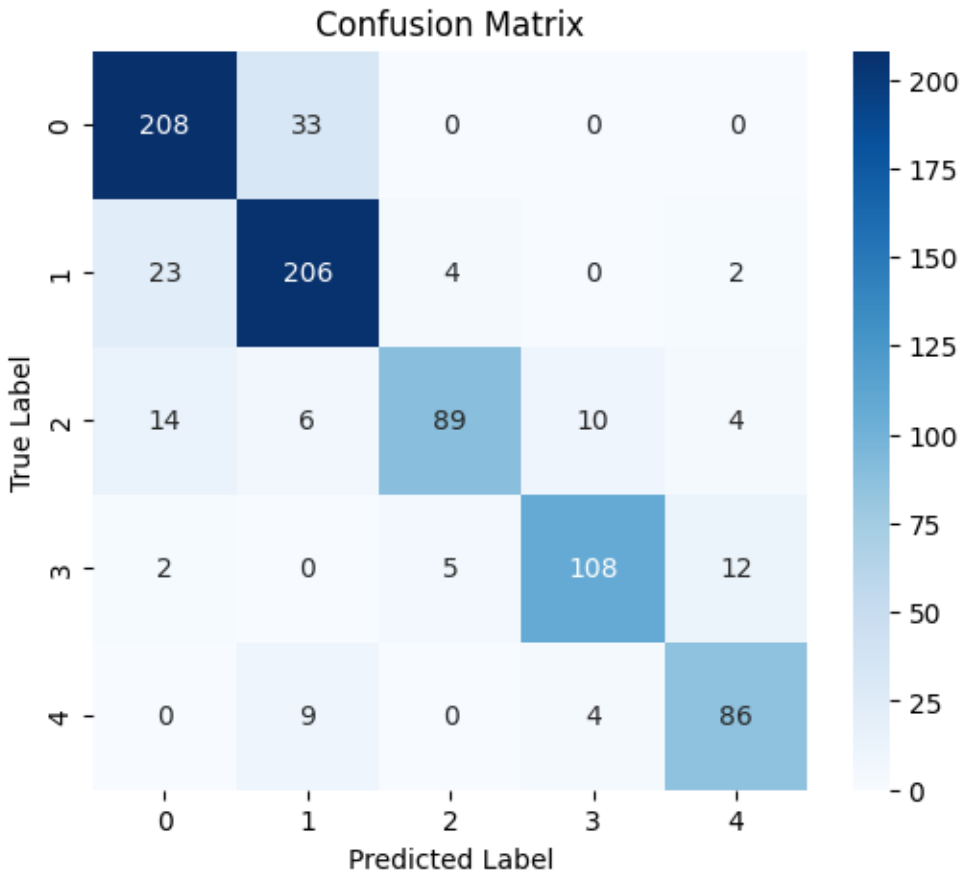

Fig.6

DENSENET

	precision	recall	f1-score	support
0	0.7539	0.9300	0.8328	257
1	0.6780	0.6911	0.6845	259
2	0.6333	0.3115	0.4176	122
3	0.8854	0.7944	0.8374	107
4	0.7955	0.8750	0.8333	80
accuracy			0.7406	825
macro avg	0.7492	0.7204	0.7211	825
weighted avg	0.7334	0.7406	0.7255	825

Fig.7

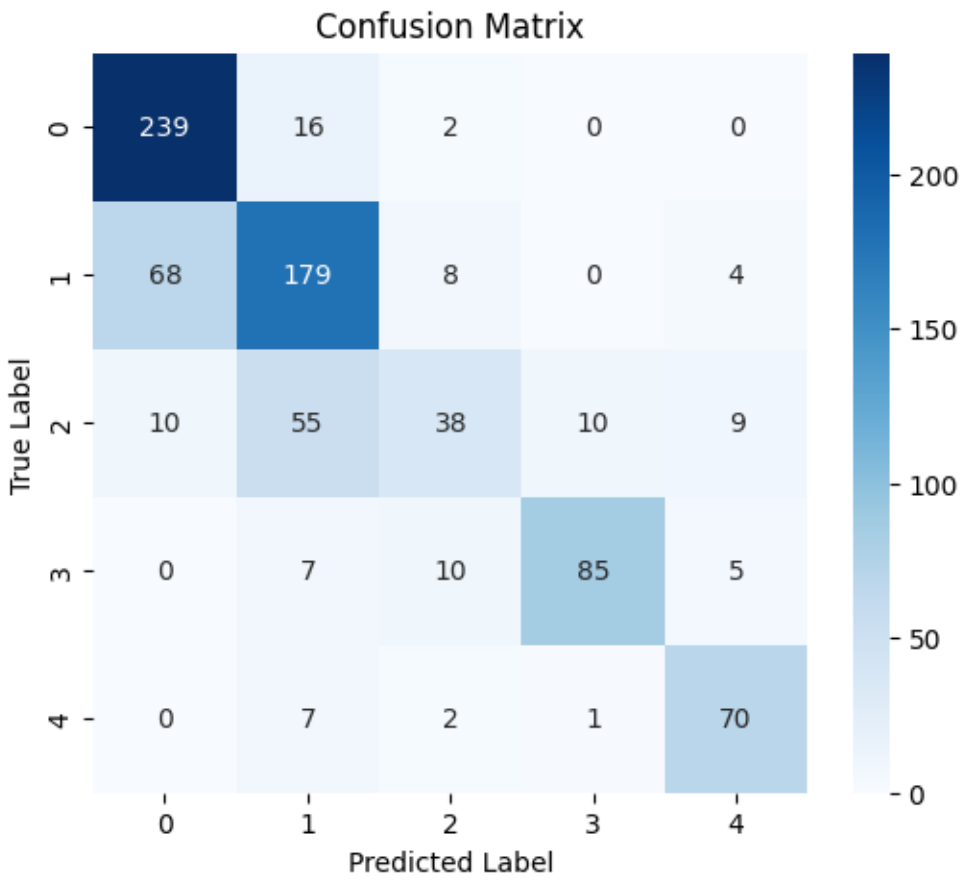

Fig.8

EFFICIENTNET

	precision	recall	f1-score	support
0	0.85	0.92	0.89	268
1	0.72	0.76	0.74	214
2	0.79	0.45	0.57	118
3	0.86	0.79	0.82	120
4	0.70	0.89	0.78	105
accuracy			0.79	825
macro avg	0.78	0.76	0.76	825
weighted avg	0.79	0.79	0.78	825

Fig.9

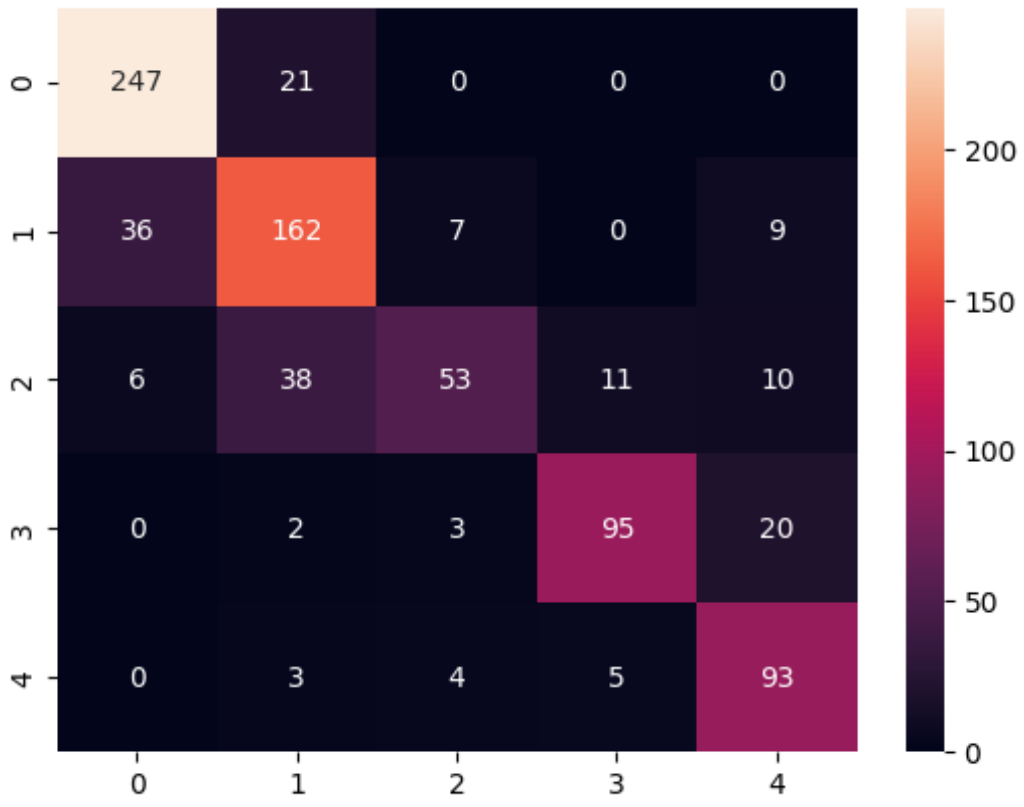

Fig.10

Web Application

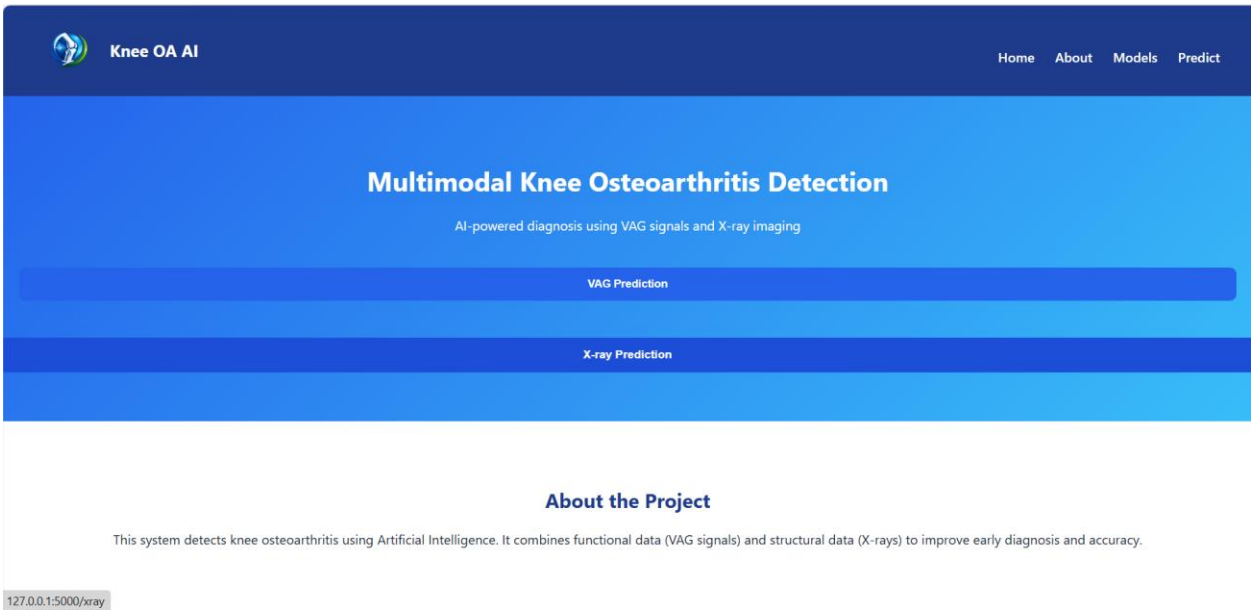

Fig.11

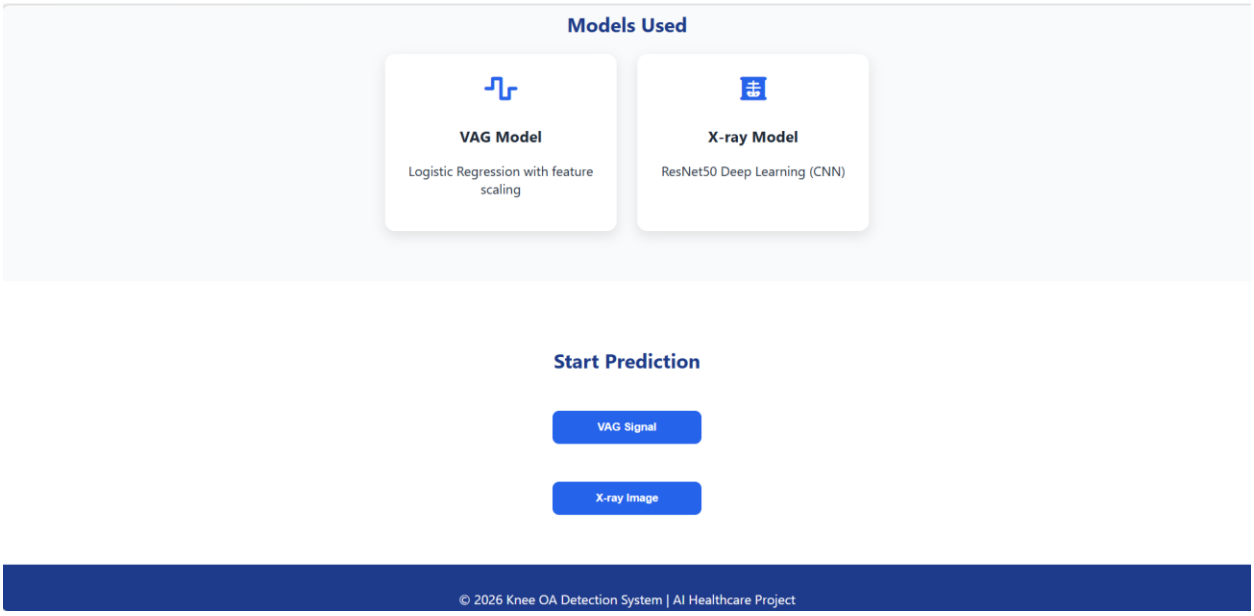

Fig.12

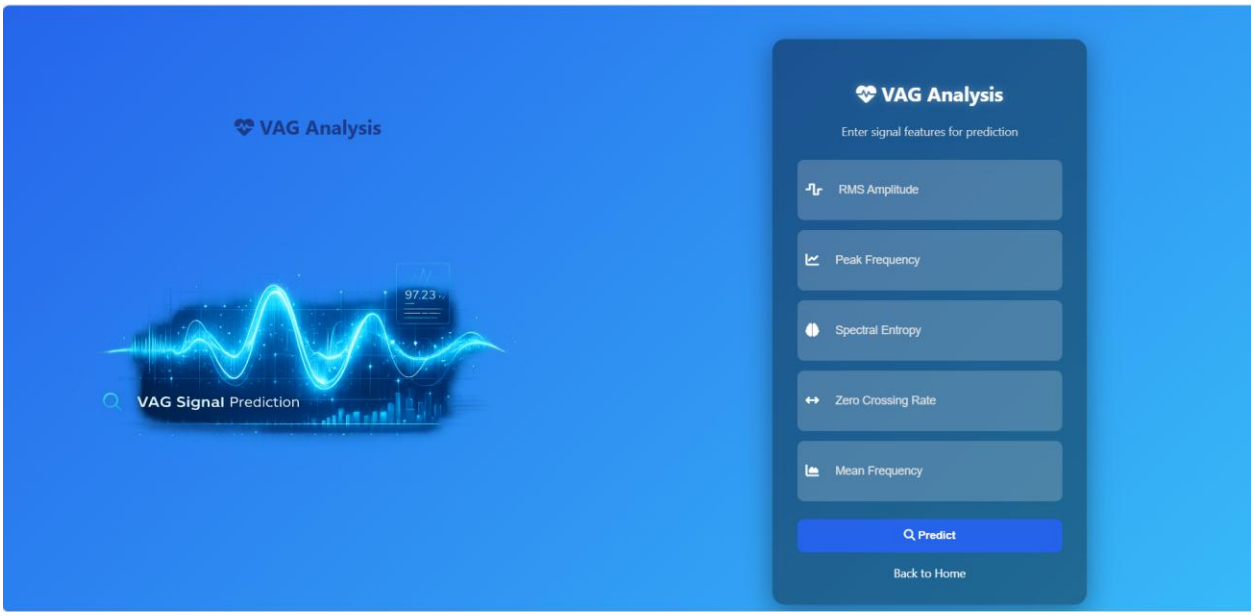
VAG Analysis
97.23
VAG Signal Prediction
VAG Analysis
Enter signal features for prediction
RMS Amplitude
Peak Frequency
Spectral Entropy
Zero Crossing Rate
Mean Frequency
Predict
Back to Home

Fig.13

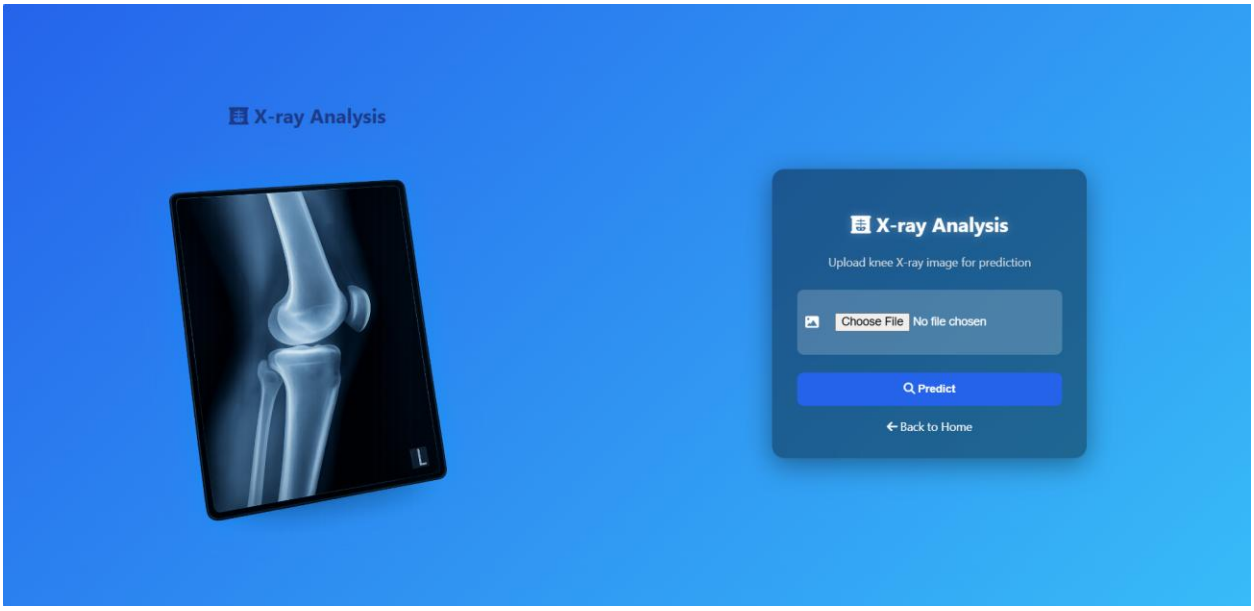
X-ray Analysis
L
X-ray Analysis
Upload knee X-ray image for prediction
Choose File
No file chosen
Predict
Back to Home

Fig.14

11. Learning Outcomes

Upon the successful completion of this project, the learner will be able to:

1. Understand End-to-End Multimodal System Design

• Gain practical knowledge of designing a complete AI-based healthcare system integrating signal processing, image analysis, and decision-level fusion.

• Understand how multiple modules (VAG signal analysis, X-ray classification, and fusion) interact within a unified architecture.

2. Apply Machine Learning for Biomedical Signal Analysis

• Learn how to use machine learning models such as SVM and Random Forest for classifying knee conditions using VAG signal features.

• Understand feature extraction, scaling, and handling of biomedical datasets.

3. Implement Deep Learning for Medical Image Classification

• Gain experience in using CNN architectures such as ResNet50, DenseNet121, EfficientNet-B0, and custom CNN models.

• Understand how to perform image preprocessing, augmentation, and transfer learning for improved accuracy.

4. Perform Multimodal Data Integration

• Learn how to combine heterogeneous data sources (signal and image) using decision-level fusion techniques.

• Understand the importance of integrating functional and structural data for better diagnosis.

5. Develop Data Preprocessing and Feature Engineering Skills

• Gain knowledge in preprocessing techniques such as normalization, scaling, encoding, and image resizing.

• Learn how proper preprocessing improves model performance and accuracy.

6. Evaluate and Optimize AI Models

• Understand how to evaluate models using metrics like accuracy, precision, recall, F1-score, and confusion matrix.

• Learn to compare different models and optimize performance through tuning.

7. Design User Interface for AI Systems

• Develop skills in creating a simple and user-friendly interface for uploading data and displaying results.

• Learn how to present outputs such as condition, severity, and recommendations clearly.

8. Strengthen Software Engineering & Documentation Skills

• Gain experience in writing system documentation including requirements, architecture diagrams, flowcharts, and feasibility analysis.

• Learn modular coding practices and system design principles.

9. Work with Real-World Healthcare Data

• Learn to handle structured (VAG features) and unstructured (X-ray images) datasets.

• Understand challenges in medical data processing and model generalization.

10. Build a Scalable Healthcare AI System

• Gain experience in designing a scalable system that can be extended to real-time monitoring and telemedicine applications.

• Learn how the system can evolve into a full-scale healthcare decision support platform.

Overall Learning Impact

By completing this project, the learner gains strong knowledge in machine learning, deep learning, biomedical signal processing, and medical image analysis. The project enhances problem-solving skills and prepares the learner for real-world applications in AI-driven healthcare systems, software development, and data science domains.

12. Conclusion

The proposed Multimodal Knee Health Assessment System successfully integrates vibroarthrographic (VAG) signal analysis and X-ray image classification to provide a comprehensive diagnosis of knee conditions and osteoarthritis severity. The system demonstrates the effectiveness of combining functional and structural data using machine learning and deep learning techniques.

In Module 1, various models such as SVM, Random Forest, Logistic Regression and CNN were used to classify knee conditions based on VAG signal features. In Module 2, multiple deep learning architectures including ResNet50, DenseNet121, EfficientNet-B0, and a custom CNN were implemented for X-ray based severity classification.

Among all the models, **ResNet50 achieved the highest accuracy of 93.3%**, **logistic Regression 88.6%** outperforming other architectures. This highlights the effectiveness of transfer learning and deep feature extraction in medical image analysis. The results confirm that deep learning models, particularly ResNet50, are highly suitable for osteoarthritis severity classification.

The modules through decision-level fusion further enhances diagnostic reliability. Overall, the system provides a cost-effective, non-invasive, and scalable solution for early detection and severity assessment of knee osteoarthritis, with strong potential for future development in real-world healthcare applications.

13. Future Scope

The proposed Multimodal Knee Health Assessment System provides a strong foundation for further enhancements and real-world applications. The system can be extended and improved in several ways to increase its effectiveness, scalability, and clinical relevance.

1. Real-Time VAG Signal Acquisition

• Integrate wearable sensors to capture VAG signals in real-time instead of relying on pre-recorded datasets.

• Enable continuous monitoring of knee health for early detection and preventive care.

2. Clinical Validation and Deployment

• Validate the system using real patient data in collaboration with hospitals and healthcare institutions.

• Deploy the system in clinical environments for practical use and performance evaluation.

3. Advanced Explainable AI Integration

• Incorporate explainable AI techniques such as Grad-CAM, SHAP, or LIME to provide visual explanations of model predictions.

• Improve trust and transparency for medical professionals.

4. Mobile and Web Application Development

• Develop a fully functional mobile or web-based application for easy access by doctors and patients.

• Enable remote diagnosis and telemedicine support.

5. Expansion to Other Joint Disorders

• Extend the system to analyse other joints such as hip, shoulder, or elbow conditions.

• Increase the applicability of the system in orthopaedic diagnostics.

6. Integration with Electronic Health Records (EHR)

• Connect the system with hospital databases and EHR systems for patient history tracking.
• Enable long-term monitoring and personalized healthcare insights.

7. Improved Multimodal Fusion Techniques

• Explore advanced fusion methods such as feature-level fusion and attention-based models.
• Enhance overall prediction accuracy and robustness.

8. AI-Based Treatment Recommendation System

• Extend the system to suggest treatment plans, physiotherapy exercises, or preventive measures based on severity level.
• Provide complete decision-support for healthcare professionals.

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15. Artefacts in the Project

Artifacts represent the key outputs, documents, and components developed during the project lifecycle.

1. Data Artifacts

• **VAG Dataset (CSV File):**

Contains extracted signal features such as RMS amplitude, spectral entropy, peak frequency, zero-crossing rate, and mean frequency used for classification.

• **X-ray Image Dataset:**

Medical images categorized based on Kellgren–Lawrence grades (0–4) used for severity classification.

• **Pre-processed Datasets:**

Cleaned, normalized, and scaled versions of datasets used for model training and evaluation.

2. Model Artifacts

• **Trained Machine Learning Models:**

SVM and Random Forest models,cnn, logistic regression trained for VAG signal classification.

• **Deep Learning Models:**

CNN model for VAG data and advanced models such as CNN, ResNet50, DenseNet121, and EfficientNet-B0 for X-ray classification.

• **Saved Model Files:**

Models saved in .pkl (ML models) and .keras formats for reuse and deployment.

3. Code Artifacts

• **Data Preprocessing Scripts:**

Python scripts for cleaning, scaling, encoding, and preparing datasets.

• **Model Training Scripts:**

Implementation of ML and DL models using Scikit-learn, TensorFlow, and Keras.

• **Evaluation Scripts:**

Code for generating accuracy scores, confusion matrices, and performance metrics.

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4. Design Artifacts

• **Flowcharts:**

Step-by-step process from input to output.

• **Architecture Diagram:**

Layered structure of system components.

5. User Interface Artifacts

• **UI Prototype Screens:**

Screens for data upload, processing, and result display.

• **Output Display Screens:**

Final results showing knee condition, severity level, and recommendations

6. Documentation Artifacts

• **Project Report:**

Complete documentation including introduction, methodology, results, and conclusion.

• **Requirement Analysis Document:**

Functional and non-functional requirements.

• **Feasibility Study & Risk Analysis:**

Technical, economic, and operational feasibility details.

• **User Manual:**

Instructions for using the system.

7. Testing Artifacts

• **Test Cases:**

Defined scenarios for validating system functionality.

• **Performance Results:**

Accuracy, precision, recall, and confusion matrix outputs.

• **Validation Reports:**

Model comparison and performance analysis.

8. Output Artifacts

• **Prediction Results:**

Outputs showing knee condition and severity classification.

• **Model Performance Graphs:**

Accuracy/loss curves and confusion matrices.

9. Deployment Artifacts (Optional)

• **Executable Scripts / Notebooks:**

Final runnable code for demonstration.

• **Saved Models for Inference:**

Ready-to-use models for prediction.

Summary

The artifacts collectively demonstrate the complete development lifecycle of the project, including data preparation, model development, system design, implementation, testing, and final output generation. These artifacts ensure reproducibility, validation, and proper documentation of the system.

16] Links

https://github.com/shravanigurram/Major-Project-2026